

UAV path planning based on K-means algorithm and simulated annealing algorithm

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Abstract: This paper proposed a new method for unmanned aerial vehicle (UAV) path planning based on K-means algorithm and simulated annealing (SA) algorithm, which solves the problem for multi-UAVs with multi-mission under complicated constraints. Firstly, the model is established for the no-fly zone, the target zone and the valid zone for cruise within it in the mission area. Then, the decomposition technique decomposes the valid area into multiple sub-target points reasonably. Secondly, the K-means algorithm is used to cluster the target points of UAV cruise, which solves the problem for UAV cruise range and scheduling issues. Combining the SA algorithm for the similar sub-target route planning, this technique increases the coverage of the UAVs in the sub-target area of cruise valid area. Finally, taking the real data of UAVs in earthquake relief as an example, the effectiveness and robustness of the proposed method is validated by simulation experiments.

Key Words: UAV; model area decomposition technique; K-means algorithm; SA algorithm

1 Introduction

In recent years, with the rapid development and application of UAV. Many researchers have noticed that it will perform very well in war and national disasters. Our country is unceasingly expanding the application of UAV. Such as electronic warfare, communications relay, air defense suppression and reconnaissance, and early warning and anti-air strikes^[1]. Path planning has become a basic precondition for ensuring UAVs successful completion of its mission. There are many environmental constraints around the UAV when it comes to missions. Such as quantity, coverage, obstacles, battery life and so on. So, in order to make the UAV can successfully complete the task. Need to consider the issue of the number of UAVs, each UAV cruise range, the UAV optimal cruise planning in cruise target area, and UAV cruise coverage issues.

For the above problems, many scholars have proposed some track planning algorithms. Such as the use of hierarchical decomposition. Multi-UAV path planning based on Voronoi diagram^[2-5]. The Probabilistic Road Map (PRM) method can also be used to study the path planning problem for multi-UAV to complete a task together^[6]. In addition, for the number of UAV arrangements, researchers use the clustering algorithm to cluster the target points. After the similar target points are clustered, they are divided into the same set by rules. Currently, countries already have a lot of clustering algorithms, such as hierarchical clustering algorithm, SOM neural network clustering algorithm, FCM fuzzy clustering algorithm. In [7], SOM neural network clustering algorithm was used to classify the target data, so the regions with denser target points were obtained. In [8], using fuzzy clustering method to solve the cloud computing under the unreasonable distribution of large data and poor efficiency issues. Although many researches have been done on data clustering and UAV path planning, most of the existing algorithms are based on the study of single point

clustering or path planning. When the UAV cruising area is large, the existing algorithms cannot realize the UAV quantity arrangement, each UAV cruise range arrangement and the UAV cruise path planning, that is, the UAV cannot implement the overall track planning.

Based on previous researchers, in this paper, the optimal planning of UAV optimal trajectory under the three constraints of UAV quantity assignment, obstacle constraints and effective cruise coverage in irregular regions is considered. Firstly, the whole model is established by using the simulation equivalent terrain method, and then the model decomposition technique is used to decompose multiple large target regions into multiple sub-target points. Secondly, according to the requirements of UAV, the target points are classified by K-means algorithm. Finally, the simulated annealing algorithm is used to map out a shortest path for all the target points that the UAV will cruise, and maximize the UAV cruise coverage. The innovation of this article has the following three points: First, in terms of the model, this paper uses the proposed model decomposition technique to decompose the large target area into multiple small target points. Second, this article treats many small target points as different states. Combining the k-means clustering algorithm and the simulated annealing algorithm, the cruise range is reasonably arranged for each UAV and the optimal flight path is planned so that the shortest time and the maximum coverage under the premise of full coverage sub-area in the process are obtained. Thirdly, through the three-step planning and design, the integrity of the navigation path of the UAV is realized, so that the UAV can be detached from the remote control to realize intelligence.

2 UAV Path Planning Problem Analysis and Model Establishment

2.1 Description of the UAV path planning problem

A number of UAVs work together, and path planning determines the number of UAVs, cruise range, cruise time and safety. At present, there are some problems in UAV path

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planning technology. Such as difficult to realize flexible autonomous control, separation of task management and control, multiple planning constraints, low intelligence and lack of multi-stage integrity for flight mission. UAV path planning constraints can generally be divided into two types. The first is the UAV flight environment constraints, constraints are generally: high altitude, fixed or moving obstacles, cruise coverage; The second is the UAV's own performance constraints, the constraints are generally: UAV maximum horizontal turning angle, maximum climb / pitch angle, life time, speed and minimum flight altitude^[9]. This paper completes the corresponding path planning under the coexistence of these two types of constraints. The trajectory planning problem considered in this paper is described as follows: According to the given number of UAVs and the life time, there are obstacles and the total cruising area has irregular environment. It is required to arrange a reasonable trajectory for each UAV under the conditions that meet the requirements, Coverage is as high as possible for all UAV cruise ranges compared to the total cruise area. The above constraints to meet the requirements are:

1. The duration of the UAV limits its number. If the subject knows the number of UAVs. The number of UAVs determines the number of total cruise zones. In each UAV cruise area, the selection of the target point determines the trajectory point.

2. The cruising area of each UAV is irregular. In order to make the cruise coverage as high as possible. This path needs to be satisfied: it is optimized under the measure of a certain performance index to ensure the minimum cost.

First of all, using the model decomposition technique, a number of target points are planned according to certain rules in the total cruising area.

The first part, based on the number of UAVs, uses the k-means algorithm to find out the target points planned in the total cruise area. In the second part, the simulated annealing algorithm is used to track the path of each UAV cruise area. This article uses multiple UAVs to work together. When the UAVs complete their mission together, their path planning can be summarized as a kind of optimization problem^[10].

2.2 Model Construction of UAV Trajectory Planning

When a UAV performs its mission in a complex environment, it needs to solve different problems at different stages. For example, a UAV is based on multiple target points for path planning, so the total cruise area needs to be broken down into multiple target points. Target selection rules, which determine the coverage of the UAV cruise. Therefore, a good model building is also one of the determinants of optimization.

In this paper, by using the equivalent terrain simulation method and using the idea of data and map fusion, a terrain information model is constructed by fusing terrain and constraints of the flight region, which simplifies the application of the algorithm. UAV track planning is divided into two phases, namely: the first stage is to establish a basic digital map model, through the three-dimensional coordinates and the UAV flight height limit to create a map of different altitudes; In the second phase, the total irregular cruising area of UAV is divided into some target points

according to rules. The article follows a UAV track planning as an example.

Stage 1: The model sets the high-altitude area as a UAV no-fly zone and defines the altitude at which the mission requires UAV inspections as an active or feasible zone. The two regions are marked with different colors in the model, in order to distinguish between different altitudes.

$Z(x_i, y_j)$ is defined as the altitude corresponding to the horizontal coordinate point x_i, y_j is the horizontal and vertical coordinates of a point on the ground level projection ($x_i=0, a, 2a, 3a \sim M-1; y_j=0, a, 2a, 3a \sim N-1; i=0, 1, 2, 3 \sim M-1; j=0, 1, 2, 3 \sim N-1$). a is a parameter. Take a coordinate point every fixed distance a in the x or y axis direction. H_0 is a reference terrain height for determining whether the cruise area is an effective area. He is the flight-able altitude of the UAV (assuming that the height of the UAV from the ground is greater than or equal to the safety altitude). H_s is the altitude of no-fly zone. Definitions A, B, C, D, E, F ... are symbols for multiple target areas. The coverage ratio in this paper is determined by the ratio of H_0 to the total target area. If the height of the UAV is one of the threats, this paper establishes a target fitness function F , which can calculate the optimal departure altitude Hi of the UAV. F is described as:

$$F = \sum_{i=1}^M \left(\frac{\sqrt{(x_i - x_{i-1})^2 + (y_j - y_{j-1})^2 + (z_{(i,j)} - z_{(i-1,j-1)})^2} + \omega \cdot \frac{\sqrt{(z_{(i,j)} - H_1)}}{2}}{\omega} \right) \quad (1)$$

where ω is the weight coefficient that depends on the track's tendency to avoid or follow the terrain.

Stage 2: Due to the limited cruise angle of the UAV and the irregular topography. The UAV's view angle is very small compared to the total cruise area due to the constraints of the UAV itself. The UAV needs to operate at each target point to maximize the cruise coverage ratio.

In response to this problem, the model decomposes the effective area. According to the rules, the effective area is divided into several target points as the target points of the UAV track planning, and then the algorithm is optimized to make the UAV cruise according to a certain route and finally reach the optimal path. Therefore, the model construction makes the number of UAV, track, cruise coverage and so on to achieve the best. The cruise area decomposition rules in this paper are described as follows:

Step 1: After extracting the coordinate points of the total cruise area, it is stored in the matrix $S_{n \times m}$, where the n -th row and the m -th column of S correspond to the altitude at the coordinate point. The first coordinate point in the top left of the cruise area is defined as (x_p, y_p, z_p) .

Step 2: Divide the area where the altitude is less than H_0 in area E. In the E area take $x_p \sim x_s$ on the abscissa. UAV elevation angle of more than J . According to the relation of J, H, T , calculate the maximum radius R of the projected ground when the UAV is flying at the same altitude, and draw the center E_0 of the first sub-region after decomposition. $E_0 = (x_0, y_0, z_0)$,

$$E_0 = \sqrt{r_0^2 - (x - x_p)^2 - (y - y_p)^2} + z_p$$

E_0 is a three-dimensional planar hemisphere. The projection level is a circle.

Step 3: E_0 is the starting point of the UAV, each point of the X-axis coordinate length is $2r_0$. $E_l=(x_{p+l}, y_{p+l})$, $x_{p+l}=x_p+2r_0$, $y_{p+l}=y_p+2r_0$. Then $x_p \sim x_s$ is decomposed according to this rule.

Step 4: In the effective area, each point of the Y-axis coordinate length is $2r_0$. The same rules as Step 2, Step 3.

Step 5: Repeat Step 2, Step 3, Step 4 until the irregular effective area is completely decomposed.

Following the above model decomposition steps, the UAV cruise area is divided into many target points. Marked as the target point S, after the decomposition is completed, the UAV according to the target point to the new path most planning. The model decomposition technique is proposed, which improves the accuracy of the latter algorithm. The UAV path will be simpler and will improve the effectiveness of the UAV as a whole rate.

3 UAV path planning based on K-means algorithm and simulated annealing algorithm

3.1 Optimization model construction

Due to the influence of complex constraints, the visible area of the UAV flight is much smaller than the effective target area. If the UAV patrol cruises on a sub-target point, resource wasting caused by the UAV in any direction will be avoided. Therefore, an optimized model building is a prerequisite to improve the optimization of later algorithms and the efficiency of UAV. In this paper, the cruise area of the model is decomposed. That is, the cruise area of the UAV is decomposed into many small areas. The center of each area is the target point of the UAV cruise. This article has explained in detail the decomposition of the region in Section 2.2 Model Building. First, in order to maximize the coverage ratio of UAV cruises, a model decomposition technique is used. Secondly, the k-means algorithm is used to classify the target points. The number of clusters is related to the number of UAVs. Finally, the simulated annealing algorithm is used to track the UAV in each area, which makes the UAV path the best and the shortest time.

3.2 Target points gathered into a class of algorithms

When some UAVs collaborate on a cruise mission, the randomness of UAV path can lead to cruise repetition and resource waste. In order to avoid low cruise efficiency and low coverage of UAV, this paper uses K-means algorithm to cluster the target points in the whole cruise area according to the number of UAV, and the number of clusters is also K. After the target points are clustered, the k-means algorithm divides the whole cruise area into K small cruise areas. Each UAV cruises a zone. When the UAV is flying at a constant altitude, the UAV has a fixed safety distance λ from the ground. So, no-fly zone Altitude: $H_s = H_c - \lambda$. The coordinates of the region within the altitude can determine whether the non-fly zone, $F(x_i, y_i)$ is defined as:

$$F(x_i, y_i) = \begin{cases} Z(x_i, y_i) \geq H_s, & \text{Non-fly zone} \\ Z(x_i, y_i) < H_s, & \text{Fly zone} \end{cases} \quad (2)$$

The height of the cruise zone is lower than the flight altitude H_c of the UAV. The height of the target point after the decomposition is also within the non-no-fly zone.

For the target point clustering, this paper uses k-means algorithm. K-means^[11], also known as K-means, is a cluster-based clustering algorithm. In this algorithm, Sum of the Squared Error (SSE) is used as an objective function to measure the quality of clustering. In the K-means algorithm, the similarity between the data is measured by the Euclidean distance. The smaller the distance between data, the more similar, the denser the data distribution, the more likely to form clusters. The average of these dense area data is usually selected as the K-means algorithm to randomly select the initial cluster centers. The number of central points is determined according to the number of clusters^[12]. The number of clusters in this paper is the number of UAVs.

The distance matrix $D=(d_{ij})_{n \times m}$ is defined according to the target point, where d_{ij} represents the distance between two target points, where D is a real symmetric matrix. In this paper, the k-means algorithm is used to cluster the target points after decomposition, as follows:

Step 1: Randomly select K (where K is the number of UAV) elements from the target points as seed points, which are regarded as the respective centers of K classes. The seed points are labeled $A, B, C \dots$ and the total number is K.

Step 2: Find the distance of K seed points for all points, and the distance is also called the degree of dissimilarity to K cluster centers. If the target point S_i is closest to the seed point A, then S_i belongs to the point A population, that is, the target point elements are respectively assigned to the cluster with the lowest degree of dissimilarity.

Step 3: According to the clustering result, recalculate the center of each K cluster, and calculate the arithmetic average of the dimensions of all the elements in the cluster^[13].

Step 4: All the elements in the target point are re-clustered according to the new center point.

Step 5: Repeat Step 2, Step 3, and Step 4 until the clustering result no longer changes.

Step 6: The result is output. Each type of target point is a sub-area of the total area, and K areas are respectively denoted as A area and B area.

The above steps are k-means clustering of the target points in the UAV cruise area. After clustering the target points into K classes, the total cruise area is divided into K parts. As a result, a large number of UAVs are routinely scheduled to cruise each area separately.

3.3 Path planning algorithm

Each target area after clustering is ranked UAV for cruise. Simulated annealing algorithm for UAV track optimal planning. The principle stems from the annealing of solid matter in physics. Usually, when solid material is annealed, it is warmed up to allow its particles to move freely, and then the particle system is brought down slowly enough that the rate is slow enough. The system is approximately at the thermodynamic equilibrium point, and finally the particle system will reach its lowest energy state, that is, the ground state, which corresponds to the global minimum point of the energy function. The objective function of the optimization problem is equivalent to the energy, the optimal solution is

equivalent to the lowest energy state. The simulated annealing algorithm randomly changes from one state to another state at a given temperature, and uses a random acceptance criterion to judge. When the temperature slowly drops to a very low value, it stays at the optimal solution with probability 1.

Taking the E area as an example, the area below the altitude H_0 in the E area is the valid area of the UAV cruise calculation coverage. The effective area generally has the characteristics of irregular size, irregular edges and different altitude. The effective area after the decomposition of the target points were defined as $A_0, A_1, A_2 \dots A_n$. Where A_0 is the start of the UAV cruise and A_n is the cruise end. Taking into account the UAV's overall cruising time requirement, the UAV will cruise from A_0 to A_n if taken as an end point. An improved simulated annealing algorithm can be used to plan the optimal trajectory to ensure the shortest time and the maximum coverage.

According to each coordinate point and altitude, the problem translates into a shortest path starting from point A_0 , cruising all the target points and reaching the end point A_n . Algorithm design process is as follows:

Step 1: Define the solution space S as a set $A_0, A_1, A_2 \dots A_n$, an element fixed starting point A_0 to end point A_n arrangement $S = \{(\pi_{A_0}, \pi_{A_1}, \pi_{A_2} \dots \pi_{A_n}), (\pi_{A_0}, \pi_{A_1}, \pi_{A_2} \dots \pi_{A_n})\}$ is a circular arrangement of $(A_0, A_1, A_2 \dots A_n)$. Each circulation arrangement represents a loop of drone cruising through all the sub-areas. $\pi_{A_i} = \pi_{A_j}$ means that at the i -th cruise sub-target point j , the initial solution is $(\pi_{A_1}, \pi_{A_2} \dots \pi_{A_n})$. Using the Monte Carlo method as the initial solution.

Step 2: According to Step 1, Iterate to create a new solution for the new track. For example, an iterative solution is: $\pi_{A_0}, \pi_{A_1}, \pi_{A_2} \dots \pi_{A_{u-1}}, \pi_{A_u} \dots \pi_{A_{v-1}}, \pi_{A_v} \dots \pi_{A_n}$

Then exchange the order between u and v to produce a new solution.

Step 3: The objective function of this phase design cost function f is the path length of all sub-regions after the UAV cruise decomposition, defined as:

$$\min f(\pi_{A_0}, \pi_{A_1}, \pi_{A_2} \dots \pi_{A_n}) = \sum_{i=A_0}^{A_n} d_{\pi_{A_i} \pi_{A_{i-1}}} \quad (3)$$

According to the data of distance matrix $D = (d_{ij})_{n \times m}$, the Metropolis criterion tests the solution to the cost function difference:

$$\Delta f = \left(d_{\pi_{A_{u-1}} \pi_{A_v}} + d_{\pi_{A_u} \pi_{A_{v+1}}} \right) - \left(d_{\pi_{A_{u-1}} \pi_{A_u}} + d_{\pi_{A_v} \pi_{A_{v+1}}} \right) \quad (4)$$

Step 4: According to the difference between the cost function to determine whether to accept the new path planning. Algorithm to determine whether to accept the new path rules are as follows:

$$P = \begin{cases} 1, \Delta f < 0 \\ \exp(-\Delta f / T), \Delta f \geq 0 \end{cases} \quad (5)$$

If $\Delta f < 0$, a new path is received; otherwise, a new path is accepted as judged by probability $\exp(-\Delta f / T)$, that is, a random number $rand$ uniformly distributed in $[0, 1]$ interval is generated by computer, and if $rand$ is less than or equal to $\exp(-\Delta f / T)$, new solution is accepted.

Step 5: Set the cooling process control parameters, the initial temperature T_0 , Coefficient of cooling coefficient α , termination temperature S_f , the current temperature iteration times U . Take the new temperature as $\alpha T_i = T_{i+1}$, select $\alpha = 0.999$. If the temperature is T_i , u times the random cycle. At this temperature, each molecule already has enough opportunity to reach the best position so the temperature can be lowered to αT_i . If the temperature T_i , the successful solution to improve L times. This is equivalent to having enough molecules in the annealing process to reach the best position. At this time, the temperature is also lowered to αT_i . The algorithm stop rule is that when the temperature drops to the end temperature S_f , the algorithm stops, which means that the minimum temperature has been reached.

Step 6: Use the selected end temperature $e = 10^{-30}$ to determine whether the annealing process is over. If $T < e$, then the algorithm is over, output the current state and solution, the algorithm is terminated.

According to the above steps, simulated annealing algorithm is used on the target points decomposed by the model decomposition technique to map out the optimal path of UAV cruising over all points. So as to achieve the shortest UAV cruise phase, the maximum coverage, the overall combination of multiple target areas, multi-UAV formation team can achieve the optimal coordination of the task planning.

4 Simulation example

4.1 Example Introduction

In this paper, we use model simplification, K-means clustering algorithm and simulated annealing algorithm for the overall track planning of UAV. In order to verify the rapid convergence in the overall situation, the multi-UAV overall track planning with the shortest time, the optimal coverage characteristics, this section simulates the proposed model decomposition technique and two algorithms.

In this paper, the application of UAV in earthquake disaster reduction and disaster reduction in a disaster area is selected as an example. By using the real coordinates and altitude data of the disaster area, a model is established and a K-means algorithm and a simulated annealing algorithm are used to coordinate the overall track of the UAV.

This example knows the region coordinates and altitude data matrix $D = (d_{ij})_{n \times m}$, which has 2913 columns and 2775 rows. The first column of the first row indicates the altitude $Z(0,0)$ at the horizontal coordinate $(0,0)$ and the distance between the selected points is 38.2 meters. That is, the data of row i and column j is expressed as $(38.2(i-1), 38.2(j-1))$.

The UAV Cruise Zone is a zone below a fixed altitude of H_0 , defined as $H_0 = 3$ km above sea level, below which H_0 is the main cruise target zone. The cruising effective area that satisfies this requirement is much larger than the visible area of the UAV and the cruising area has an irregular boundary state.

The UAV used searches for signs of life with a life detector, which provides accurate target positioning for post-disaster relief. The number of UAVs $K = 30$, they depart from the base stations $H(110,0)$ and $J(110,55)$ (in kilometers) and return to their respective places of origin

upon completion of the mission. Detector effective detection range of not more than 1000 meters. The average flight speed is 60 km / h. The UAV elevation angle from the ground is 60° . The UAV cruises below H_0 for a safe flight of 50 meters and UAV for 8 hours. Set UAV flight altitude $H_e = 4200$ meters, these UAVs are required to these areas UAV 3 kilometers below the region can be detected by the detector as much as possible, and from the first UAV fly to the last base time interval Try to be short.

4.2 Model Simplification and Clustering Algorithm Simulation

For the actual data in Section 4.1, the equivalent model of the simulated map is shown in Fig. 1. The right side of the figure color scale unit: m. The number corresponds to the altitude of the horizontal coordinate of the map. Different altitudes are identified by different colors in the figure.

Within the total area, the UAV cruise area determined by H_0 shows an irregular topography. The coverage of the UAV cruise to this effective area is determined by the cruising trajectory of the UAV in this area. According to the model decomposition technique Step1 ~ Step5 introduced in Section 2.2 of this paper, the total area is decomposed into target points as shown in Fig. 2. Fig. 2 is a top view of Figure 1. Upon request, the simulation simplifies the model by reducing the part of the terrain above 3000 meters above sea level. In Fig. 2, the blue circle is the simulation result of Step 1 ~ Step5 of Section 2.2 model decomposition. The irregular part composed of the blue part is the area below 3000 meters in the whole terrain.

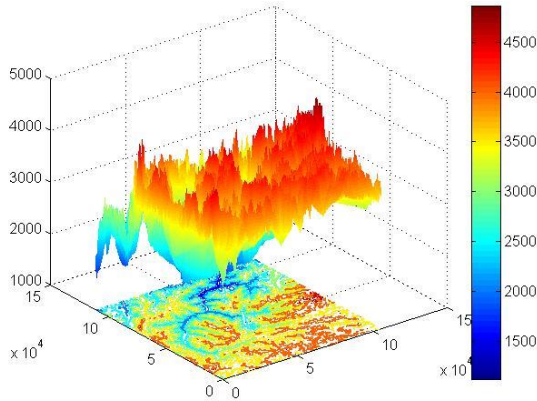


Fig. 1: Simulation map equivalent model diagram

In the total area, the UAV cruise area determined by the H_0 presents an irregular terrain state, and the UAV inspection of the effective area coverage is determined by the UAV's cruise trajectory in the area. According to the model decomposition techniques introduced in section 2.2 of this paper, Step1~Step5 show that the total area is decomposed into target points. FIG. 2 is a plan view of FIG. According to the requirement, the model is simplified by the part of the terrain below the altitude of 3000 meters. In Fig. 2, the blue circle is the simulation results of the 2.2-step model decomposition technique Step1~Step5. The irregular area formed by the blue part is the area where the elevation is less than 3000 meters in the whole terrain. That is, Fig. 1 shows

the three-dimensional terrain simulation in the example, and Fig. 2 shows the altitude of the area below 3000 meters in the top view of Fig. 1. The region is then subjected to decomposition and k-means clustering algorithm simulation. The result is the blue in Fig. 2. The area represented by the color.

According to the k-means clustering algorithm Step1 ~ Step6 introduced in Section 3.2 of this article. Clustering the target points that were decomposed before. The number of cluster center points K is the number of UAVs 30. The black symbol in Fig. 2 is k-means clustering algorithm simulation results.

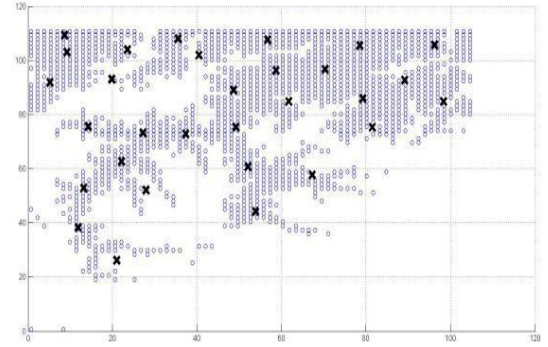


Fig. 2: Model simplification and clustering algorithm simulation map is decomposed into target point map

According to the elevation angle of the UAV, the altitude of the ground and the turning radius, the radius of the decomposition sub-area is taken as $r_0 = 700\text{m}$. When r_0 is taken as the UAV, it is at the ground view angle range.

4.3 Simulated Annealing Algorithm Path Planning Simulation

The last section clustered the target points, the purpose of which is to reasonably distribute the cruising area of each UAV. Target point clustering classifies the total cruise area so that each UAV cruise does not interfere with each other and saves time and resources. This section uses the simulated annealing algorithm in Section 3.3 of this paper to plan the path of the target points in each area after clustering. Take the classified partial area as an example to plan the trajectory of each UAV. From the starting point H or J point to the center of the area, a simulated annealing algorithm is used. Part of the path planning diagram is shown in Fig.3.

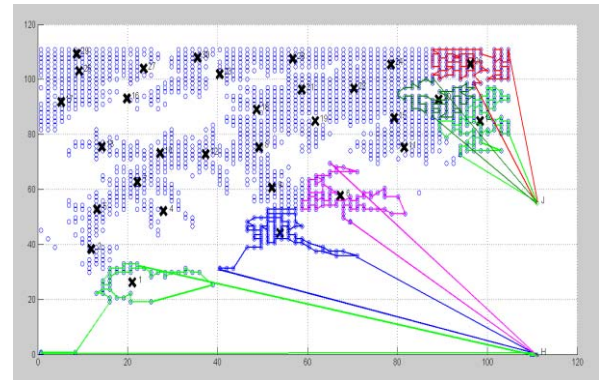


Fig. 3: Simulated Annealing Algorithm Part Path Planning

The lower right hand position H base station in Fig. 3, the upper base H is the J base station. The 30 UAVs start at points H and J, respectively, and the six UAV cruise routes are shown in different colors in the figure. The algorithm terminates the loop at any time in the case of local minutiae, so that the final solution tends to be globally optimal. The simulated annealing algorithm determines whether the new state solution is accepted by probability, that is, when the initial temperature is sufficiently high, the temperature is slow enough, the sample is long enough and the final temperature approaches zero, the algorithm eventually converges to the optimal solution with the probability 1.

The simulated annealing algorithm used in this stage is higher than the Hop-field neural network algorithm, genetic algorithm, etc., and the simulated annealing algorithm has a high execution rate and success rate. The simulated annealing algorithm incorporates decomposition techniques and is suitable for path planning in very large areas, and can simplify large-area cruise efficiency and improve cruise coverage. Hop-field neural network algorithm is suitable for path planning of very few target points, and more target points will not be able to plan paths. Genetic algorithm converges quickly but easily falls into local optimum. It is difficult to jump out of the local optimum and cause oscillation after late convergence. The improved simulated annealing will not fall into the local optimum and late oscillations, making the UAV cruise the shortest time.

The above simulation results show that although the environment constraints and the constraints of the UAV itself are complex, the model decomposition technique, the k-means clustering algorithm and the simulated annealing algorithm are adopted in the simulation, so that the UAV efficiently completes the task required by the example. Due to the large number of UAVs with high coverage cruise to multiple target areas, this paper designs cruise coverage of more than 92% in effective area and satisfies the requirements of UAV cruising during the cruising time. In this paper, the model of the target area was initially established, which greatly simplified the difficulty of the UAV cruising in the target area, and effectively improved the accuracy, optimization and coverage of the algorithm planning path, and practically solved the problem of resource waste.

5 Conclusion

In this paper, UAV is studied in the aspects of model building, clustering of target points in UAV cruise area and trajectory planning of UAV to achieve the best coverage. Then proposed model decomposition technology, k-means clustering algorithm and simulated annealing algorithm. It comprehensively solves the UAV scheduling assignment, flight avoidance and path planning coordination, cruise coverage and other issues in complex constraints. While establishing the optimal mathematical model, the model is effectively decomposed and improved so that various problems of UAV cruising are effectively simplified before the algorithm is applied, so that the accuracy of the improved algorithm is optimized. The effectiveness of optimization

models and algorithms is verified by examples, which improves the overall UAV efficiency and cruise coverage. At the same time, the overall design of this paper can meet the requirements of multi-UAV with optimal track, highest coverage and regular UAV formation cruise from base station to target area. It can accomplish the task under complex constraints and make UAV controllable from man-made control autonomous navigation.

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